Introduction

Homelessness is a looming issue for many developed countries, and it is important to gauge the public's perception of the problem. Do some countries villainize the homeless population? Do others view the issue with optimism and work towards solutions? The cultural differences between countries and how they view this issue provide interesting insights into localized societal norms and, potentially, into predictions of future trends in homelessness. To address this topic, this study used national news headlines from four countries (the United States, the U.K., Australia, and Canada) and conducted sentiment and framing analyses of news titles related to homelessness. Four English-speaking countries were chosen to use the same English-reading models and avoid potential translation errors.

Methods

The collection and analysis process can be broken down into five steps as follows.

1. Data Collection

Article titles were collected from a Media Cloud API across four national press collections: the United States, the United Kingdom, Canada, and Australia. Data spans from 2014 to 2024. Articles were filtered by the following Boolean query:

homelessness OR homeless OR "rough sleeper" OR "rough sleepers" OR "rough sleeping" OR unhoused OR unsheltered OR "housing insecurity" OR "housing insecure" OR "tent city" OR "tent cities" OR encampment

5,000 titles per country per year were retrieved, with an equal number (416 to 417) per month to achieve a more representative sample. In total, 220,000 articles were collected. For each article, the title, URL, publication date, and source outlet were collected. All analyses were conducted on headline text only.

2. Data Cleaning

Duplicate URLs were removed, duplicate title strings within the same country were removed, and blank/missing titles were also removed. This process reduced the sample from 220,000 to 86,144 articles. This significant drop-off acts as a limitation of this study, and were this work to be reproduced, an initial filter to prevent blank or duplicate titles would be implemented in the data collection phase.

3. Sentiment Analysis

Two complementary methods were utilized for sentiment analysis.

VADER: Each title received a compound score in [−1.0, +1.0], classified as positive (≥ +0.05), negative (≤ −0.05), or neutral (between).

RoBERTa (cardiffnlp/twitter-roberta-base-sentiment-latest): a model trained on ~124 million tweets. It outputs three-class probabilities (positive, neutral, negative), with the dominant class taken as the sentiment label and a compound score derived as positive_probability − negative_probability.

4. Framing Analysis

Seven different framing themes were assessed in two different formats (keyword matching and zero-shot classification). The seven frames were: Criminalization**,** Humanization**,** Policy / Government**,** Health / Services**,** Crisis / Alarm**,** Dehumanizing Language, and Empathetic Language**.** Frame prevalence was expressed as the percentage of titles per country and year that matched each frame.

Keyword matching applied case-insensitive whole-word regular expression matching against manually curated keyword lists for each frame. The manually crafted word lists for each of the seven themes are in the framing_keyword.py file. Zero-shot classification used cross-encoder/nli-MiniLM2-L6-H768, a cross-encoder fine-tuned on the MNLI corpus, with each frame represented as a plain-language hypothesis (e.g. *"policing, arrests, or criminalization of homeless people"*). Titles with an entailment score ≥ 0.50 were flagged as matching that frame. Both methods were applied independently and compared side-by-side to assess convergence.

5. Coverage Analysis

To provide additional context for the report, an analysis was conducted of the proportion of all articles published in each country that addressed homelessness. Results were assessed by the proportion of all articles in each national collection that matched the homelessness query, calculated per country per year. This data was collected from the Media Cloud API.

Results

Sentiment (Vader & RoBERTa)

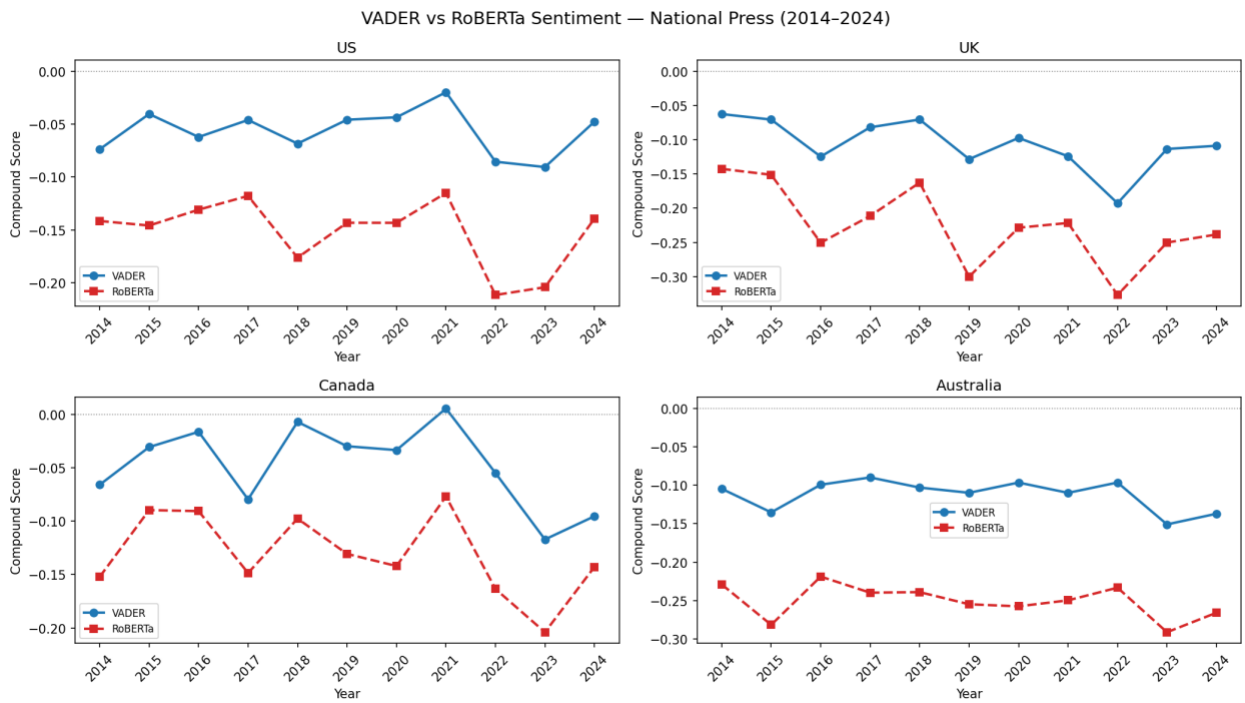

Figure 1:VADER and RoBERTa Sentiment across National Press in Selected Countries (2014-2024)

Sentiment analysis using both VADER and RoBERTa found that homelessness coverage was consistently negatively toned across all four countries throughout the 2014–2024 period, with no country averaging above neutral in any year. Across all panels, RoBERTa consistently produced more negative compound scores than VADER, typically by 0.10 to 0.20 points, suggesting that the transformer-based model is more sensitive to the negative framing in headline language. Despite this gap in magnitude, the two methods follow a strikingly similar directional pattern, attributing more confidence to the findings.

Canada showed the least negative sentiment under VADER, approaching neutrality in several years (2018, 2021), while the UK and Australia registered the most consistently negative scores under both methods. Notably, all four countries show a pronounced dip toward more negative sentiment in the 2022–2023 period under RoBERTa, which may reflect heightened coverage and negative reactions to post-pandemic homelessness spikes. By 2024, scores in most countries show a partial recovery toward less negative levels, though coverage remains consistently below neutral across both methods.

Framing (Keyword & Zero-Shot)

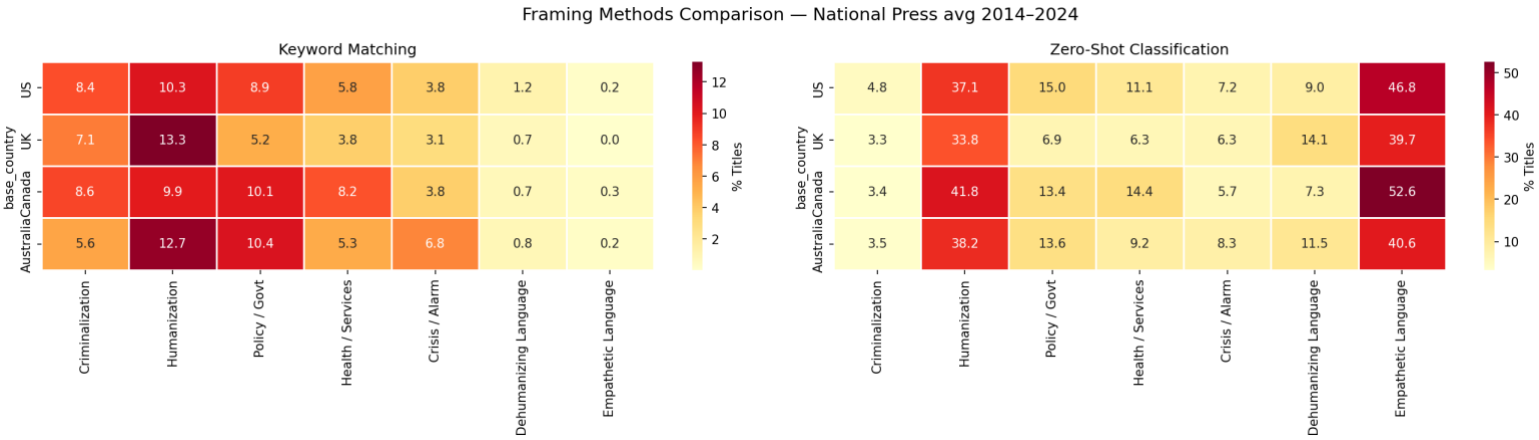

Figure 2: Heatmap of Keyword and Zero-Shot Framing Methods Results for National Press of Four Countries on Homelessness (2014-2024)

Both keyword matching and zero-shot classification identify Humanization as the dominant frame across all four countries, though the methods differ substantially in estimating its prevalence. Keyword matching yields humanization estimates between 9.9% (Canada) and 13.3% (UK), while zero-shot classification yields far higher estimates, ranging from 33.8% (UK) to 41.8% (Canada). A similar pattern holds for Empathetic Language, which has a keyword matching rate consistently near zero (max of 0.3%), while zero-shot classification identifies it as the second most prevalent frame, ranging from 39.7% (UK) to 52.6% (Canada). This divergence likely reflects the zero-shot model's ability to detect empathetic tone semantically, rather than solely relying on the presence of specific phrases. By contrast, Criminalization and Policy / Government show much closer agreement between the two methods, with keyword matching producing figures of 5.6%–8.6% and 5.2%–10.4%, respectively, and zero-shot producing 3.3%–4.8% and 6.9%–15.0%. Dehumanizing Language is consistently rare across both methods in all countries, most likely due to the filter that national-level news undergoes before publication.

Coverage Analysis

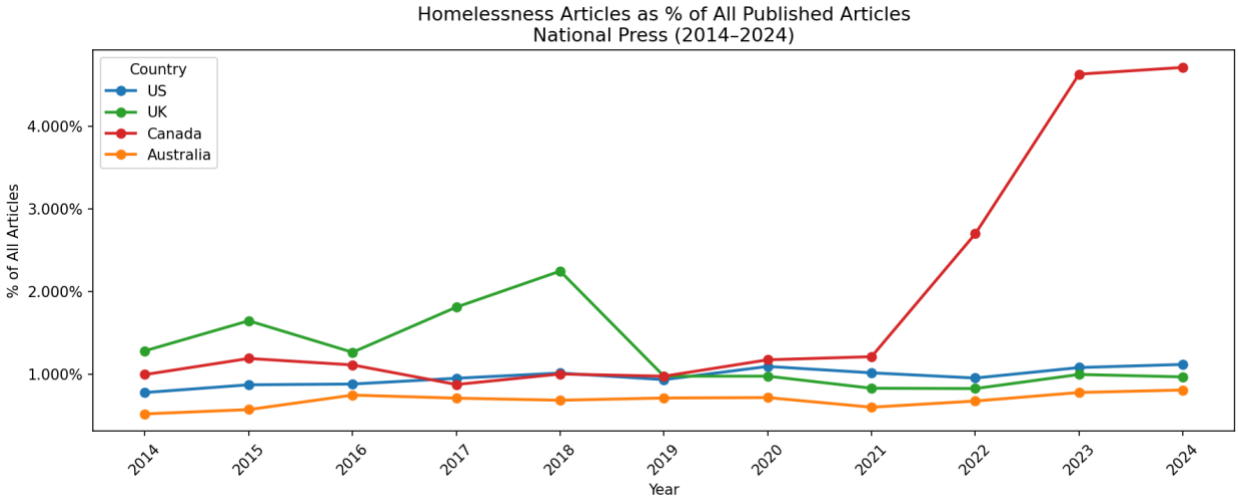

Figure 3: Homelessness Articles as % of All Published Articles in the National Press (2014-2024)

Homelessness coverage as a proportion of all published national press articles was relatively modest across all four countries throughout the study period, generally ranging from 0.5% to 1.5% for most country-year combinations. Australia consistently recorded the lowest share, rarely exceeding 1.0% in any year. The US maintained a stable and largely flat trajectory throughout, hovering around 0.8%–1.1%. The UK showed elevated coverage in the 2014–2018 period, peaking at approximately 2.2% in 2018, before declining sharply and converging with the US and Australia from 2019 onward. The most interesting pattern is Canada's dramatic surge from approximately 1.2% in 2021 to 4.6% by 2023, and it remained at that elevated level through 2024. This near quadrupling of homelessness coverage share in Canadian national press within just two years suggests a significant shift in editorial attention. This coincides with advancements in addressing homelessness in Canada during this time period, following the pandemic. Taken together, the data indicate that while homelessness coverage has remained a relatively stable minor share of national news agendas in the US, UK, and Australia, it has become a substantially more prominent issue in Canadian national press in recent years.

Takeaways

Across all four countries and both methodological approaches, this study finds that homelessness is consistently framed in national press coverage in negative terms, with humanizing and policy-oriented frames dominating. With the notable exception of Canada, countries had a relatively stable share of overall news coverage of homelessness. Sentiment analysis using both VADER and RoBERTa confirms that headline language on homelessness has remained persistently below neutral throughout 2014–2024, with no meaningful shift toward more positive or solutions-oriented coverage across countries. Framing analysis reveals that, while humanization is the most prevalent frame by volume, criminalization and crisis framing maintain a steady presence, and empathetic person-first language remains rare under the keyword-matching framework employed in this study. The substantial divergence between keyword and zero-shot methods on tone-based themes potentially highlights the limitations of keyword-based analysis methods. Canada's dramatic rise in homelessness coverage share from 2021 onward stands out as the most significant country-level finding, pointing to a fundamental shift in how prominently the issue features on the national news agenda. This shift in 2021 also points to an inflection point in the country's RoBERTa and Vader sentiment scores, as both metrics see a sharp decline (more negative) coinciding with the increase in news coverage. Overall, these findings show that across the developed, English-speaking countries' press, homelessness remains a persistently negatively framed issue, with incremental rather than structural change in how it is covered over the past decade.

Appendix

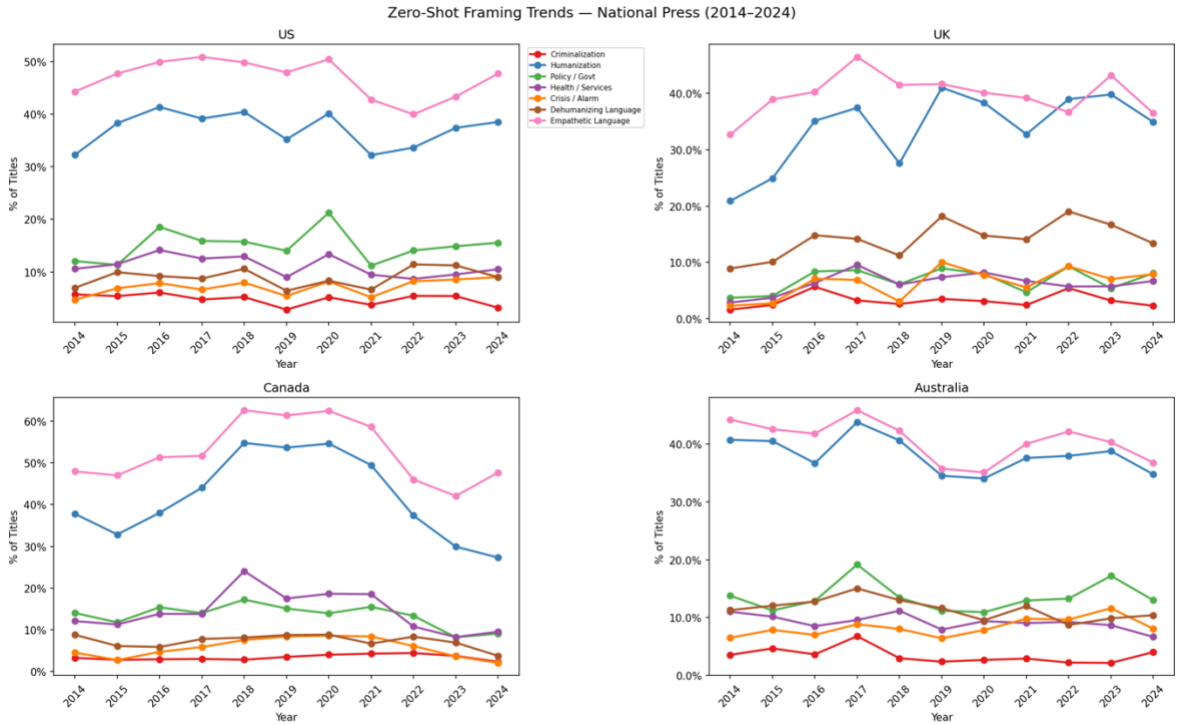

Figure 4: Full Zero-Shot Framing Analysis of National Press Across 7 Themes (2014-2024)

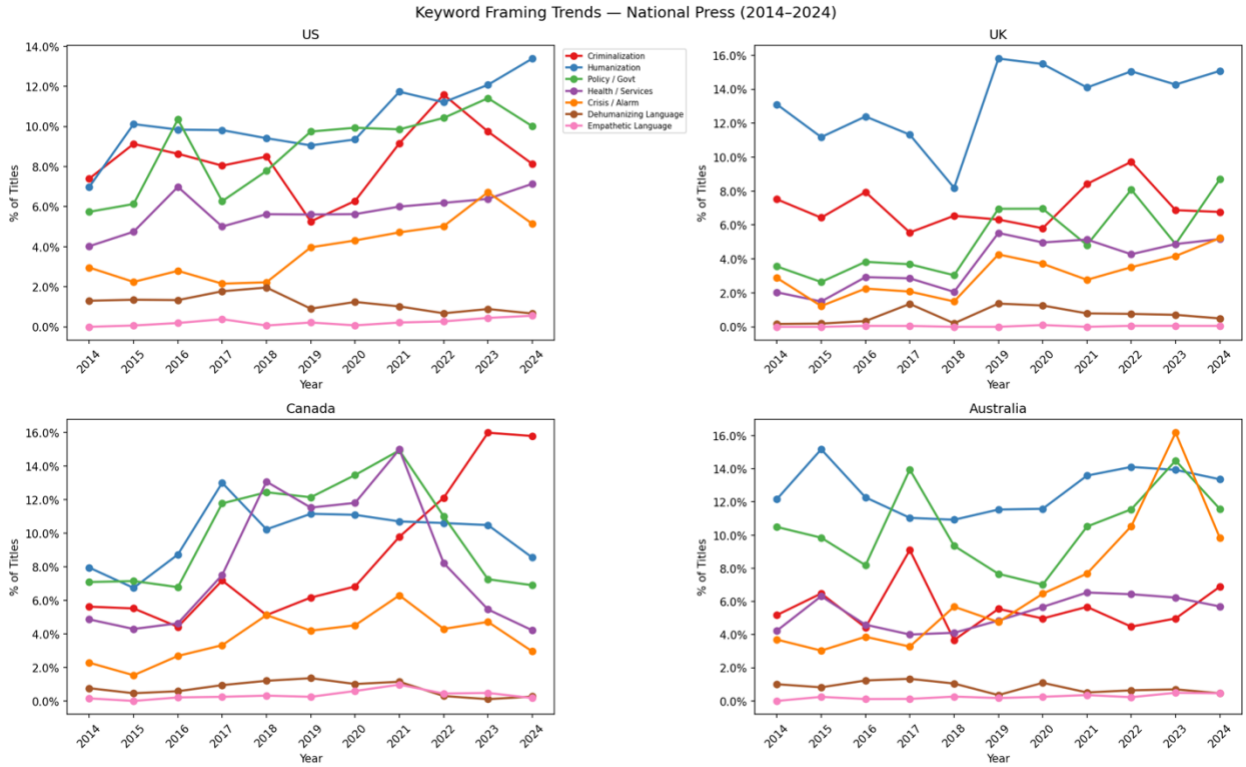

Figure 5: Full Keyword Framing Analysis of National Press Across 7 Themes (2014-2024)

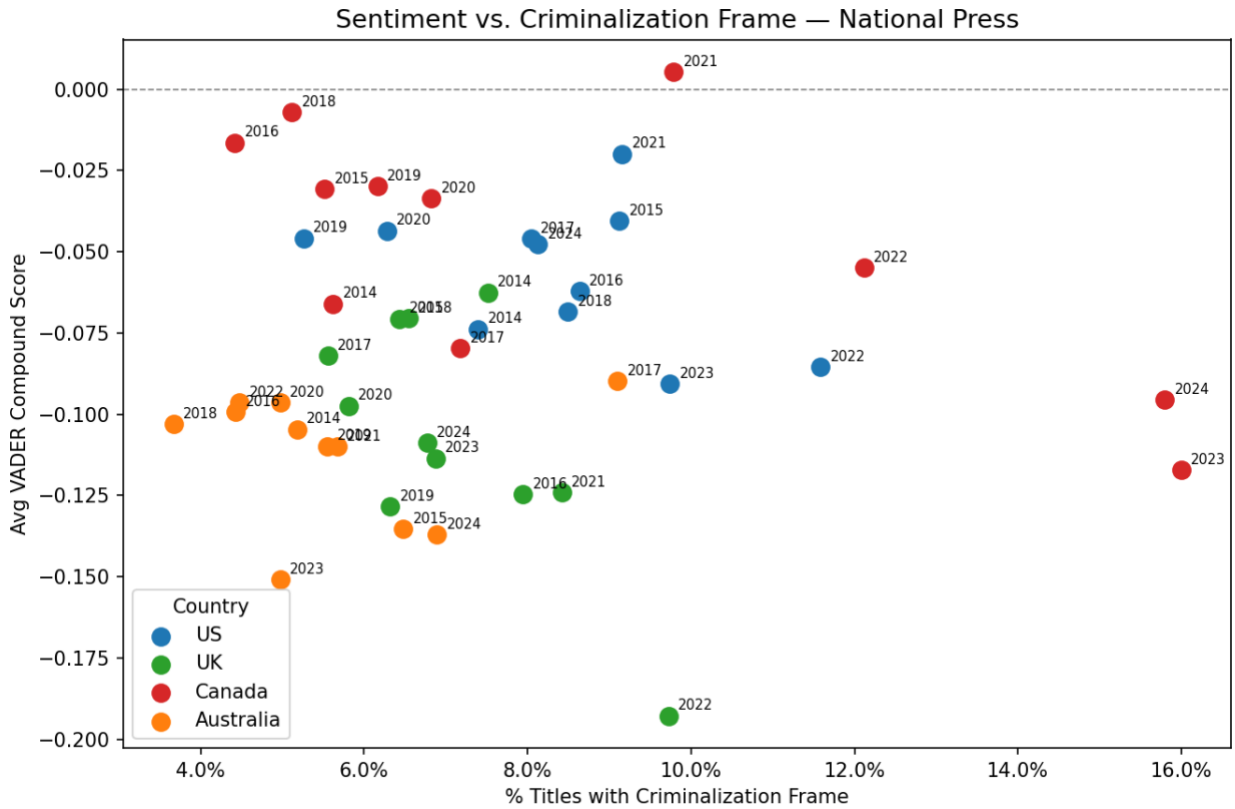

Figure 6: Relationship Between VADER Compound Scores and Zero-Shot Criminalization Rate